

Xiaoyu Zhang

CONTACT INFORMATION

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RESEARCH SUMMARY

I build **multimodal sensing agents** that *perceive, reason, and act* on the physical world by jointly leveraging sensing modalities organized across four complementary physical domains: **radio-frequency** (mmWave FMCW radar, synthetic-aperture and cascaded-array imaging), **acoustic / ultrasonic** (photoacoustic imaging, ultrasound, audio and speech), **optical / radiographic** (visible-light vision, infrared, mammography), and **bio-electrical** (wearable ear-EEG). My research closes the loop from **novel sensing front-ends** through **physics-informed inverse imaging** and **cross-modal representation learning** to **deployed clinical systems**, with validated impact on **smart health** (wound care, breast cancer subtyping, thyroid diagnosis, Alzheimer’s screening, hearing disorders) and **social good** (sustainable building, accessible HCI, point-of-care diagnostics). My work has produced **14 peer-reviewed papers** (7 first-authored, at venues including ACM IMWUT, ACM UIST, IEEE IoTJ, IEEE/RSJ IROS, and npj Imaging), a **Best Paper Candidate** at ACM SenSys Human-Sys’25, **4 patents** (one being translated through a startup I co-founded, semifinalist at Panasci TEC 2025), and a **clinical station deployed under an IRB-approved validation study**, reflecting a deliberate *research-clinic-industry* translation pipeline.

RESEARCH INTERESTS

My research lies at the intersection of **Multimodal Sensing, Embodied/AI Agents, AI for Health, and Human-Centered Computing**. The unifying question is: *how can a heterogeneous set of low-cost, non-invasive sensors be unified into an intelligent agent that reasons under occlusion, motion, and data scarcity, and acts safely in real clinical and societal settings?*

1) Multimodal Sensing Agents for Smart Health. I design end-to-end systems that fuse mmWave, photoacoustic imaging, ultrasound, ear-EEG, and speech to deliver *non-invasive, low-cost, and accessible* diagnostic and monitoring tools — from over-gauze wound assessment and photoacoustic foot-ulcer prognosis, to tri-modal breast-cancer molecular subtyping, thyroid nodule diagnosis, Alzheimer’s speech screening, and ear-EEG-based hearing-disorder triage. The agents are designed to operate where conventional imaging fails: through fabric, in low-resource clinics, and on edge hardware.

2) Embodied RF Perception for Physical and Human-Robot Interaction. I develop RF-based perception modules that endow robots and HCI systems with *contact-free, occlusion-robust* awareness of humans and materials — including pixel-level hand localization, mmWave-voice multi-factor identification, micro-expression recognition, and subsurface material reasoning for autonomous building inspection. These modules bridge wireless sensing and embodied intelligence, extending agents’ perceptual envelope beyond vision.

3) Sensing Agents for Social Good and Sustainable Practice. I translate multimodal sensing into deployed systems for societally meaningful problems: vision-driven process control for sustainable construction-material production, autonomous robotic thermal-insulation inspection for building decarbonization (IROS’26), and low-cost health monitoring for underserved populations. This thrust grounds the research in real-world stakeholders — hospitals, factories, and patients — and motivates the patent and entrepreneurship pipeline below.

EDUCATION

University at Buffalo, the State University of New York (SUNY)
Ph.D., Computer Science and Engineering

Sep. 2021 - 2026

Co-advised by Prof. Wenya Xu & Yaxiong Xie

Dissertation (**defended 2026**): *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications*
Committee: Wenyao Xu, Yaxiong Xie, Jun Xia

University of Science and Technology of China
M.S., Electronics and Communication Engineering

Sep. 2017 - Jun. 2020
Advised by Prof. Bin Liu

Hefei University of Technology
B.Eng., Electronic Information Engineering

Sep. 2013 - Sep. 2017

INDUSTRY &
CLINICAL
EXPERIENCE

Clinical Research Collaborations

2022 - Present

Multi-year IRB-approved collaborations spanning five clinical cohorts at the University at Buffalo and partner sites:

- **Vascular Surgery** — chronic foot-ulcer and wound-care cohort; on-site deployment of the WaveSight-RF clinical station
- **Breast imaging — 53-patient** tri-modal photoacoustic / ultrasound / mammography cohort for molecular subtyping
- **Thyroid** — clinical thyroid ultrasound scanning for nodule diagnosis and lesion segmentation
- **Neurology / audiology** — Alzheimer’s speech and sound-sensitivity (ear-EEG) screening cohorts
- **Photoacoustic Imaging Lab** (Dept. of Biomedical Engineering) — joint imaging-system development and radiomics analysis

Responsibilities included IRB protocol support, multi-modal dataset curation, prototype deployment in clinic, and clinician-facing decision-support design.

China Merchants Bank Software Center

Jul. 2020 - Sep. 2020

Designed and implemented a full-stack data-verification system (Bootstrap + Django) with dynamic front-back interaction for multi-source financial data.

HONORS AND
AWARDS

- **Best Paper Candidate Award**, HumanSys’25 — The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (co-located with ACM SenSys), 2025
- **Semifinalist**, Panasci Technology Entrepreneurship Competition (Panasci TEC), University at Buffalo, 2025
- **Graduate Teaching Award**, Department of CSE, University at Buffalo, 2024
- **Chair’s Fellowship**, Department of CSE, University at Buffalo, 2021
- Second-Prize Graduate Scholarship, University of Science and Technology of China (USTC), 2018, 2019
- First-Prize Graduate Scholarship, University of Science and Technology of China (USTC), 2017
- Third-class Undergraduate Scholarship, Hefei University of Technology (HFUT), 2015, 2016, 2017

PUBLICATIONS

Summary. 14 peer-reviewed papers, 7 first-authored (name in **bold**), spanning flagship venues in ubiquitous computing (ACM IMWUT, UIST), robotics (IEEE/RSJ IROS), medical imaging (Photoacoustics, npj Imaging), and biomedical informatics (IEEE JBHI, IoTJ). Thrust tags: [\[Health\]](#) [\[Embodied\]](#) [\[Social Good\]](#).

- [IROS'26] [Social Good] **Xiaoyu Zhang**, Ye Zhan, Chenhan Xu, Zhengxiong Li, Chuqin Huang, Yanda Cheng, Licheng Liang, Wei Bo, Jun Xia, Chi Zhou, Hongyue Sun, Wenyao Xu, “*Autonomous Robotic Inspection of Building Thermal Insulation via Subsurface mmWave Sensing*”, IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026. (*Patent filed.*)
- [IMWUT'26] [Health] Shuwei Hou, Wei Bo, L. Cao, Varun Shijo, Chuhui Liu, **Xiaoyu Zhang**, L. Guo, Wenyao Xu, “*Speech Annotation and Transcription Enhancer (SATE): An Automated System for Child Language Sample Analysis*”, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2026.
- [ICIBM'26] [Health] **Xiaoyu Zhang**, Wei Bo, Anarghya Das, Longxiang Pan, Wenyao Xu, “*Matching Speech Models to AD/HD Subtypes: A Mechanistic Benchmark*”, International Conference on Intelligent Biology and Medicine, 2026.
- [npj Imaging'26] [Health] Yanda Cheng, Chuqin Huang, Shuliang Yu, Saptarshi Chakraborty, Yunqi Xi, Robert W. Bing, Huijuan Zhang, **Xiaoyu Zhang**, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, “*Predictive modeling of chronic foot ulcer outcomes using longitudinal photoacoustic imaging*”, npj Imaging, 2026.
- [IoTJ'25] [Health] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, “*mmSkin: An Over-gauze Wound Assessment System using Radio Frequency Technologies*”, IEEE Internet of Things Journal, 2025. (*Translated into US patent; being commercialized through a co-founded startup.*)
- [IoTJ'25] [Embodied] **Xiaoyu Zhang**, Zhengxiong Li, Chenhan Xu, Luchuan Song, Huining Li, Hongfei Xue, Yingxiao Wu, Wenyao Xu, “*mmHand: Towards Pixel-Level-Accuracy Hand Localization Using a Single Commodity mmWave Device*”, IEEE Internet of Things Journal, 2025.
- [HumanSys'25] [Health] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, “*Through-dressing Wound Monitoring Based on the mmWave Sensor*”, The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys workshop, co-located with ACM SenSys), 2025. (**Best Paper Candidate**)
- [Sensors'25] [Health] **Xiaoyu Zhang**, Chuhui Liu, Yanda Cheng, Zhengxiong Li, Chenhan Xu, Chuqin Huang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, “*A Comprehensive Survey of Research Trends in mmWave Technologies for Medical Applications*”, Sensors, 2025.
- [Photoacoustics'25] [Health] Chuqin Huang, Yanda Cheng, **Xiaoyu Zhang**, Ye Zhan, Wenhan Zheng, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, “*Radiomics-Driven Perfusion Prediction in Clinical Photoacoustic Foot Imaging*”, Photoacoustics, 2025.
- [Photoacoustics'25] [Health] Chuqin Huang, Emily Zheng, Wenhan Zheng, Huijuan Zhang, Yanda Cheng, **Xiaoyu Zhang**, Varun Shijo, Robert W. Bing, Isabel Komornicki, Linda M. Harris, Ermelinda Bonaccio, Kazuaki Takabe, Emma Zhang, Wenyao Xu, Jun Xia, “*Enhanced clinical photoacoustic vascular imaging through a skin localization network and adaptive weighting*”, Photoacoustics, 2025.
- [JBHI'24] [Health] Wei Bo, Suzanne S. Sullivan, **Xiaoyu Zhang**, Mingchen Gao, Wenyao Xu, “*A Telemedicine Analytic Framework for Fully and Semi-automatic Alzheimer’s Disease Screening using Clock Drawing Test*”, IEEE Journal of Biomedical and Health Informatics, 2024.
- [UIST'23] [Embodied] Tiantian Liu, Feng Lin, Chao Wang, Chenhan Xu, **Xiaoyu Zhang**, Zhengxiong Li, Wenyao Xu, Ming-Chun Huang, Kui Ren, “*WaveID: Robust and Secure Multi-modal User Identification via mmWave-voice Mechanism*”, ACM Symposium on User Interface Software and Technology, San Francisco, USA, 2023.
- [BSN'19] [Embodied] **Xiaoyu Zhang**, Bin Liu, “*A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs*”, IEEE-EMBS International Conference on Body Sensor Networks, Chicago, USA, 2019.

[Bodynets'17] [Embodied] Chengjie Guan, Bin Liu, Zhiqiang Liu, Y. Zhang, **Xiaoyu Zhang**, “JMMM: A Mobility Model for WBANs Based on Human Joint Movements”, 19th EAI International Conference on Body Area Networks, Dalian, China, 2017.

COMPETITION

- **Panasci Technology Entrepreneurship Competition (Panasci TEC) 2025** Mar. 2025
Advanced Through-Gauze Wound Assessment Technology.
Xiaoyu Zhang (technical lead / co-founder), Chuhui Liu, Jun Xia, Linda M. Harris, Wen Yao Xu.
Advanced to the **semifinals**; pitched the patented mmWave wound-assessment platform to clinical and industry judges.

PRESENTATIONS & DEMOS

- **Ph.D. Dissertation Defense: Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications** 2026
University at Buffalo, CSE
- **Live Demonstration:** SUNY Research Expo 2025
WaveSight-RF clinical wound-imaging station — translational outreach
- **Device Presentation:** UB Clinical and Translational Science Institute (CTSI) Studio 2025
Through-dressing wound imaging — structured clinician-feedback session
- **Invited Talk:** The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys, co-located with ACM SenSys) May 2025
Through-dressing Wound Monitoring Based on the mmWave Sensor
- **Pitch Talk:** Panasci Technology Entrepreneurship Competition Mar. 2025
Advanced Through-Gauze Wound Assessment Technology
- **Invited Talk:** IEEE-EMBS International Conference on Body Sensor Networks May 2019
A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs
- **Invited Talk:** 19th EAI International Conference on Body Area Networks Sep. 2017
JMMM: A Mobility Model for WBANs Based on Human Joint Movements

TEACHING EXPERIENCES

Teaching Assistant across **6 semesters** at the University at Buffalo, including one invited guest lecture, plus one semester at USTC.

1. Algorithm Analysis and Design, UB Fall 2021, Spring 2022, Fall 2022, Fall 2023, Fall 2024
2. Special Topics, UB — incl. **invited guest lecture:** *Wireless Signal Processing: Making Sense of the Invisible* Fall 2023
3. Mathematical Logic and Graph Theory, USTC Fall 2018

MENTORING EXPERIENCES

I have mentored **10+ students** on wireless sensing, embedded systems, and AI-based medical screening projects.

- Prakriti Setlur — AI-based clinical image screening pipelines
- Kh Shafin Farhan — foundation models for medical imaging
- Abhi Ramtel (CSE@UB) — UI design of an over-gauze wound assessment system
- Cole Desimone (AE@UB) — 3D model design of the mmWave sensor scanning system
- George Gillman (EE@UB) — mmWave technologies for medical applications: a review
- Weida Jiang (CSE@UB) — embedded mmWave-based hand detection on Raspberry Pi
- Wenxuan Huang & Yiwen Tan (CSE@UB) — tooling for automatic image labeling
- High-school summer research interns — AI-based medical screening pipelines, mentored through UB outreach programs

Conference Organization:

- **Organizing Committee Member (Local Arrangements)**, International Conference on Intelligent Biology and Medicine (ICIBM) [2026]

Journal Reviewer:

- IEEE Transactions on Mobile Computing (TMC) [2025, 2026]
- ACM Transactions on Computing for Healthcare (HEALTH) [2025]
- IEEE Open Journal of Engineering in Medicine and Biology (OJEMB) [2025]
- Scientific Reports [2025]
- Smart Health (SH) [2024, 2025]

Conference Reviewer:

- ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) [2025]
- IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) [2024, 2026]
- Physics Education Research Conference (PERC), American Association of Physics Teachers (AAPT) [2026]
- IEEE-EMBS International Conference on Body Sensor Networks (BSN) [2023, 2024]

Outreach:

- Live technology demonstration, **SUNY Research Expo** — translational outreach for the WaveSight-RF clinical wound-imaging station [2025]
- Device presentation and clinician-feedback session, **UB Clinical and Translational Science Institute (CTSI) Studio** [2025]